

1. **Course Title:** Computational Finance
2. **Course Code:** CS 401
3. **Semester and Year:** BTech CS/ICT Sem VII
4. **Credit Structure(L-T-P-Cr):** 3-0-2-4
5. **Prerequisites** (if any): Probability and Statistics SC 215
6. **Instructor's Name with email:**
Jaideep Mulherkar
Jaideep_mulherkar@daiict.ac.in

7. **Course Objectives:**

A derivative is a financial instrument that derives its price from an underlying asset like a stock. An example is a forward contract where two parties agree to buy/sell an asset at a fixed price at some future date. Another example is that of a call option where one party (option buyer) agrees to buy an asset from another party (option seller) at a future date. But in this case the contract is asymmetric, that is, the option buyer has the right but no obligation to exercise the option whereas the option seller is obligated to exercise the option at end date. How do you value such a financial contract? The value of the option is the option price of the call option. There is a nice mathematical theory of option pricing which is based on principles of efficient markets and mathematical theory of probability. The Black-Scholes-Merton model of pricing a European call option is a landmark work in financial and economic theory which received the Noble price for economics in 1997. Current financial markets are replete with complicated financial instruments; exotic options and option exchanges are common place. Everyday billions of dollars of options are traded on various exchanges of the world. Options are traded everyday on various underlying assets like stock, currency or commodities. This course introduces the student to the theory of option pricing and financial derivatives.

8. **Suggested Textbook/references:**

- a.
- b. Options, Futures and Other Derivatives J.C. Hull and S. Basu
- c. Stochastic calculus for finance I Shreve

9. **Evaluation Scheme**

Labs and Assignments: 20%
InSem Exam : 1 Insem 35%
End Sem Exam: Final 45%

10. **Lecture Plan :**

Sr No	Description	No. of Lectures
1.	Introduce stock markets, importance of returns, information efficiency/arbitrage.	2
2.	Forward contracts and call and put options and option price determinants.	6
3.	Single period Binomial model. Equivalence between hedging, replication and risk-neutral valuation. Multi-period model.	6
4.	Random walks, scaled random walks and Geometric Brownian Motion	6
5.	Introduction to stochastic calculus:Ito's lemma Black-Scholes PDE via hedging argument and Ito's lemma Black-Scholes PDE via the risk-neutral valuation argument	4
6.	Solution to Black Scholes via Heat Equation	1
7.	Basics of Greeks and their calculation	2
8.	Monte Carlo methods for option pricing	3
9.	Finite difference methods: Explicit, Implicit and Crank-Nicholson	4
10.	American and Exotic options	3
11.	Jump diffusion model for stock pricing	1

11. Course Outcomes:

After completion of this course, students will be able to understand basics of unconstrained and constrained optimization from a theoretical and practical viewpoint.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X	X				X			X